

Modern Cloud Computing

Lecture 3: Cloud Networks

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Today's Topics

- Internet Basics
- Internet Protocols
- Named Data Networks and Software Defined Networks
- Data center Interconnection Networks
- Content Delivery Networks

Internet Basics

Networks for Cloud

Cloud computing is built on communication.

- Users access cloud resources over the Internet.
- Servers inside a cloud communicate through high-bandwidth, low-latency interconnects.
- Distributed storage, analytics, and microservices are often limited by data movement.
- Data center networks affect latency, throughput, cost, power, and reliability.

Cloud computing performance is not only about CPU and storage; it is also about the network connecting everything.

Cloud Workloads and Network Demand

Cloud workloads can be broadly grouped by their dominant resource need:

Workload Type	Network Sensitivity
CPU-intensive	Benefits from fast networking, but may not require it
Memory-intensive	Benefits from fast networking, but may not require it
I/O-intensive	Strongly depends on networking
Storage-intensive	Strongly depends on networking

Data-intensive and network-intensive applications require more expensive networks with higher bandwidth and lower latency.

Internet as Cloud Access Network

The Internet is the network used to access cloud services.

It is a **packet-switched network**:

- Data is divided into packets.
- Packets are queued and routed by switches and routers.
- Packets may experience random delay.
- Packets may be lost.
- Packets may arrive out of order.

Basic Network Terms

Term	Meaning
Datagram	Transfer unit in a packet-switched network
Header	Control information used to transport a datagram
Payload	The actual data being carried
Protocol	Rules defining how communicating parties behave
Host	End system at the network edge
Network edge	Where end-user systems reside
Network core	Routers and control systems interconnected by high-bandwidth links

Layering and Modularization

Packet-switched networks are complex systems.

Two basic design strategies:

- **Layering:** decompose communication into layers.
- **Modularization:** hide implementation details behind interfaces.

A layer usually communicates only with adjacent layers.

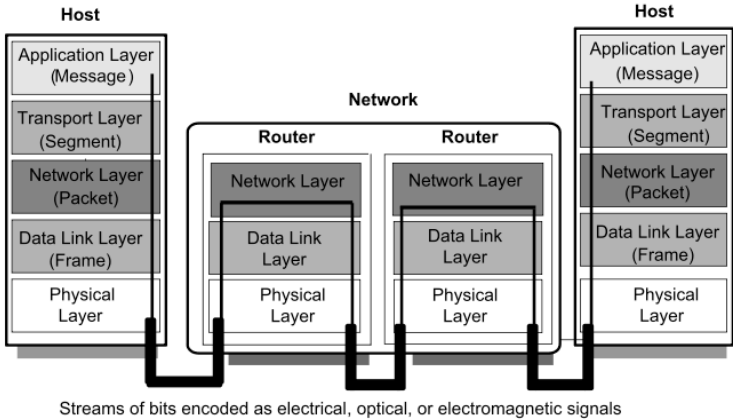
- Simplifies design.
- Allows lower layers and upper layers to evolve independently.

Internet Protocols

Internet Protocol Stack

Layer	Data Unit	Main Role
Application	Message	Application-level communication
Transport	Segment	End-to-end communication and data delivery between applications running on different host devices.
Network	Packet	Routing across networks
Data Link	Frame	Communication between adjacent nodes
Physical	Bits	Electrical, optical, or electromagnetic signals

Each layer communicates logically with its peer layer on the other host.



Encapsulation and Decapsulation

At the sender:

1. Application data moves down the protocol stack.
2. Each layer adds its own header.
3. The physical layer transmits bits.

At the receiver:

1. Bits move up the protocol stack.
2. Each layer removes and interprets its own header.
3. Payload is delivered to the application.

The Internet: A Network of Networks

The Internet is a collection of distinct autonomous networks.

All networks use:

- Globally unique IP addresses
- IP routing
- Border Gateway Protocol (BGP)

Hourglass Architecture of the Internet

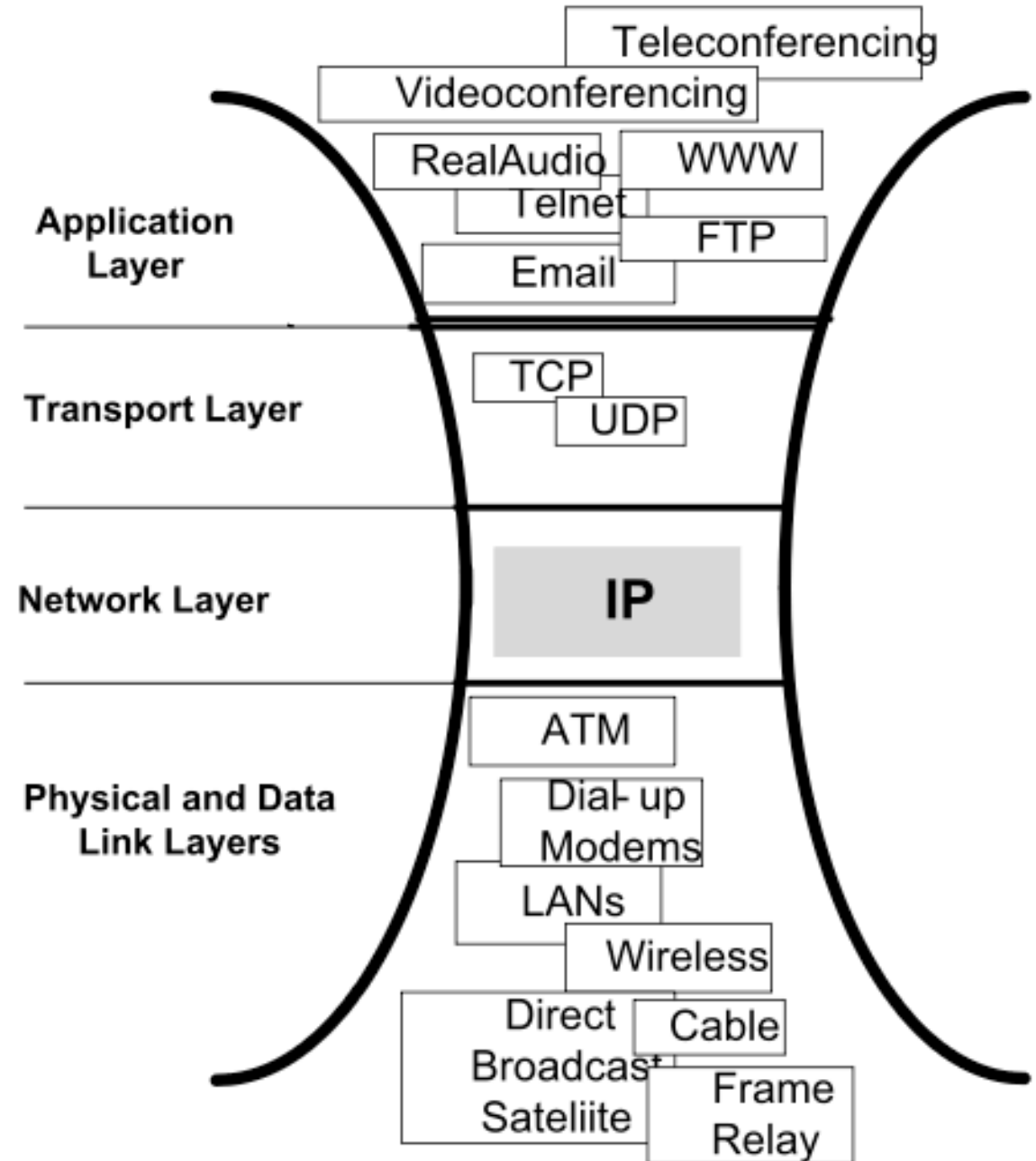
The Internet has an **hourglass architecture**.

Many applications and transport protocols sit above IP.

Many physical and link technologies sit below IP.

All packets are routed using:

- IP protocol
- Destination IP address



Best-Effort Delivery

IP provides **best-effort delivery**.

- Routers try to deliver packets.
- Routers may drop packets when overloaded.
- IP does not guarantee delivery, ordering, or latency.

Reliability, ordering, and flow control are usually handled above IP, especially by TCP.

Forwarding Plane vs. Routing Plane

The Internet separates two functions:

Plane	Responsibility
Forwarding plane	Decides what to do with packets arriving on inbound interface
Routing plane	Builds routing tables used by routers

Forwarding uses tables to map packet destination addresses to outgoing interfaces.

Routing determines what those tables should contain.

Separation allowed packet forwarding to remain stable while routing protocols evolved.

IP Address vs. MAC Address

A host usually has both logical and physical addresses.

Address	Meaning
IP address	Logical address used for routing across networks
MAC address	Physical address assigned to a network interface

IP addresses may change, especially for mobile devices.

MAC addresses are tied to network interfaces.

DHCP: Assigning IP Addresses

Dynamic Host Configuration Protocol (DHCP) automatically assigns IP addresses.

Three allocation modes:

1. **Dynamic allocation:** temporary lease from a pool.
2. **Automatic allocation:** permanent assignment from a pool.
3. **Static allocation:** address assigned based on a MAC-to-IP table.

Dynamic address assignment is common in cloud virtualized and containerized environments.

TCP vs. UDP

Feature	TCP	UDP
Type	Connection-oriented	Connectionless
Reliability	Reliable delivery	Best effort
Ordering	Ordered byte stream	No ordering guarantee
Flow control	Yes	No
Congestion control	Yes	No
Common uses	Web, email, file transfer, remote administration	DNS, VoIP, games, streaming, TFTP

TCP: Reliable Ordered Byte Stream

TCP provides reliable, ordered delivery of bytes from one application to another.

TCP uses:

- Sequence numbers
- Acknowledgments
- Timers
- Retransmissions
- Sliding-window flow control
- Congestion control

Cloud implication: TCP reliability is helpful, but its congestion-control behavior affects web and cloud application latency.

TCP Flow Control

TCP flow control prevents the sender from overwhelming the receiver.

A receiver advertises how much additional data it can accept.

The sender respects this receive window.

If the receiver advertises a window size of zero:

- Sender stops sending.
- Persist timer prevents deadlock if later window updates are lost.

TCP Congestion Control

Congestion occurs when network load approaches capacity.

Symptoms:

- Packet drops
- Increased delay
- Increased jitter

TCP tries to infer congestion using:

- Packet loss
- Round-trip time estimates
- Retransmission timeout
- Duplicate acknowledgments

Congestion Window

TCP controls the sender rate using a congestion window.

$$S = \frac{W \times MSS}{RTT}$$

where:

- S = sender data rate
- W = window size
- MSS = maximum segment size
- RTT = round-trip time

Actual window:

$$W = \min(cwnd, awnd)$$

where $cwnd$ is congestion window and $awnd$ is receiver advertised window.

TCP Slow Start and Congestion Avoidance

TCP Tahoe uses:

- **Slow start:** congestion window grows quickly at first.
- **Congestion avoidance:** window grows more conservatively after a threshold.
- **Fast retransmit:** retransmit after duplicate ACKs, without waiting for timeout.

Simplified behavior:

```
for every ACK:  
    if  $W < ssthresh$ :  $W++$   
    else:  $W += 1/W$   
  
for every loss:  
     $ssthresh = W/2$   
     $W = 1$ 
```

IPv4 Exhaustion and IPv6

IPv4 supports about:

$$2^{32} \approx 4.3 \text{ billion addresses}$$

IPv6 supports about:

$$2^{128} \approx 3.4 \times 10^{38} \text{ addresses}$$

IPv6 also changes the network architecture by emphasizing:

- Multicast rather than broadcast
- Stateless address autoconfiguration
- Integrated IPsec support

Why IPv6 Migration Is Hard

Migration to IPv6 requires updates to:

- Applications
- Hosts
- Routers
- DNS infrastructure
- Operational tools
- Security policies

Organizations also need backward compatibility with IPv4.

Changing from 10-digit phone numbers to 40-digit phone numbers would require new phones, new directories, and updates throughout the system.

NAT: Address Sharing

Network Address Translation (NAT) lets many private hosts share one public IP address.

Benefits:

- Extends lifetime of IPv4 address space
- Hides internal private addresses

Challenges:

- Can complicate peer-to-peer communication
- Historically caused problems for VoIP and VPNs
- Adds another layer of stateful network behavior

Named Data Networks and Software Defined Networks

Named Data Networks: Motivation

The Internet names communication endpoints.

But many modern applications care about content:

- Videos
- Images
- ...

Named Data Networking routes by data name instead of host address.

NDN Hourglass Architecture

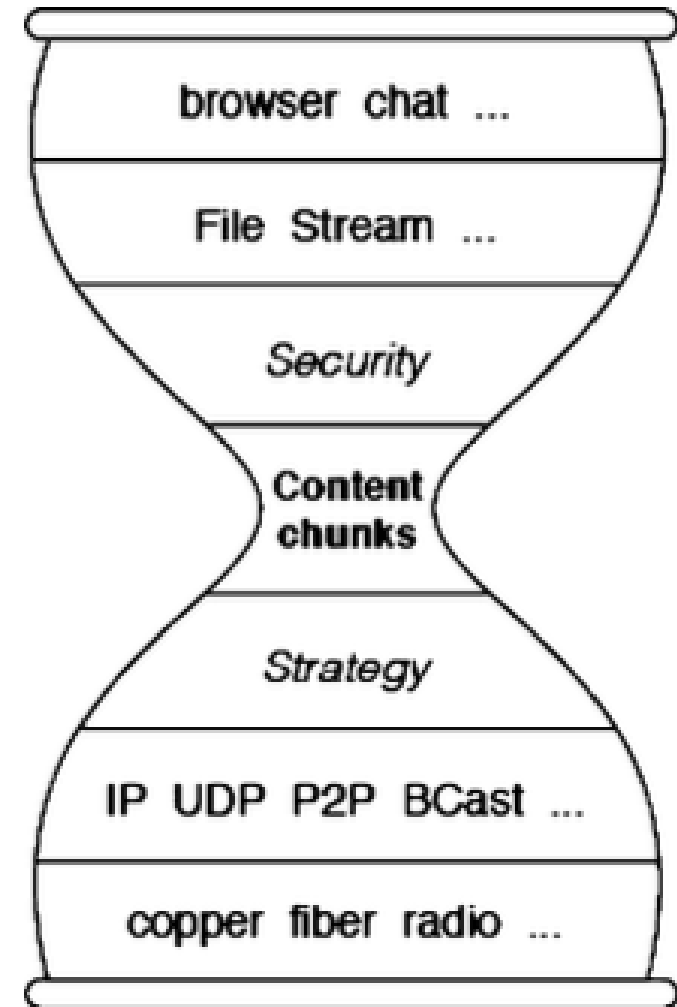
NDN keeps an hourglass-like architecture, but changes the narrow waist.

Internet narrow waist:

- IP packets routed to addresses

NDN narrow waist:

- Named data chunks fetched by name



NDN Packet Types

NDN uses two packet types:

Packet	Role
Interest	Request for named data
Data	Response carrying name, content, and signature

Process:

- 1. Consumer sends Interest packet with data name.
- 2. Router forwards Interest toward possible data source.
- 3. A node with matching data returns a Data packet.
- 4. Data follows the reverse path back to the consumer.

NDN Router State

An NDN router maintains three main structures:

Structure	Purpose
Content Store	Cache for Data packets that crossed the router
Forwarding Information Base	Name-prefix forwarding table
Pending Interest Table	Tracks forwarded Interests not yet satisfied

NDN routers can cache content and satisfy future requests directly.

NDN and Cloud Data Centers

NDN could be useful for cloud data centers when:

- Data is replicated.
- Multiple instances access the same data.
- Nearest storage server can satisfy requests.

Software Defined Networks

Software Defined Networking (SDN) applies virtualization ideas to networks.

SDN separates:

- Network configuration logic
- Physical forwarding resources

A network operating system runs in a control layer between:

- Application layer
- Infrastructure layer

Goal: allow applications or controllers to reconfigure the communication substrate programmatically.

SDN Architecture

Major SDN components:

- **Applications:** express network requirements
- **Controllers:** translate requirements into network actions
- **Datapaths:** logical network devices exposing control to the physical substrate
- **Control-to-data-plane interfaces**
- **Northbound interfaces** between applications and controllers
- **Management and administration**

OpenFlow

OpenFlow is an API for programming data-plane switches.

An OpenFlow switch has:

- Flow table
- Action associated with each flow entry
- Controller that programs the flow entries

The controller:

- Configures the switch
- Manages switch behavior
- Receives events from the switch

Cloud connection: SDN helps data centers implement flexible routing, isolation, load balancing, and policy enforcement.

Data center Interconnection Networks

Interconnection Networks for Clouds

Inside a cloud, servers communicate through interconnection networks.

A network consists of:

- Nodes
- Links or communication channels
- Switches
- Network interfaces

Interconnection networks are distinguished by:

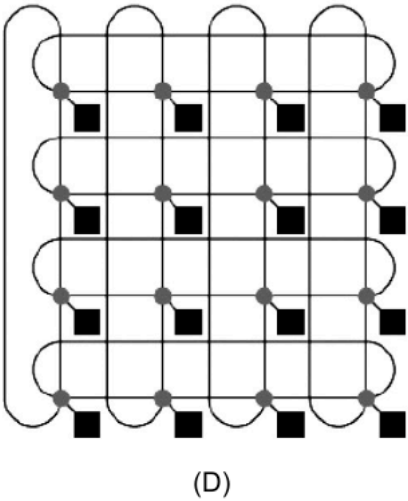
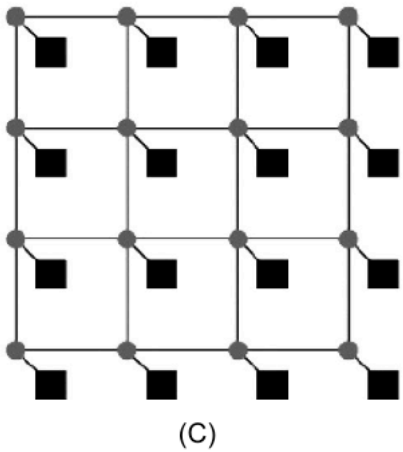
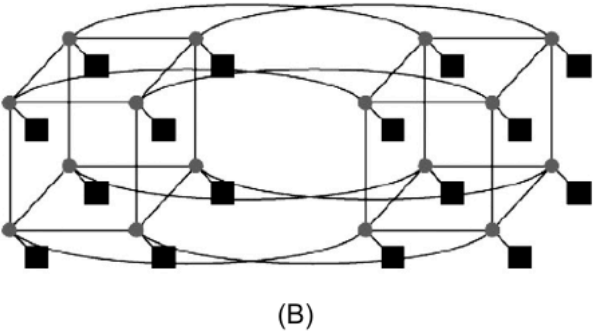
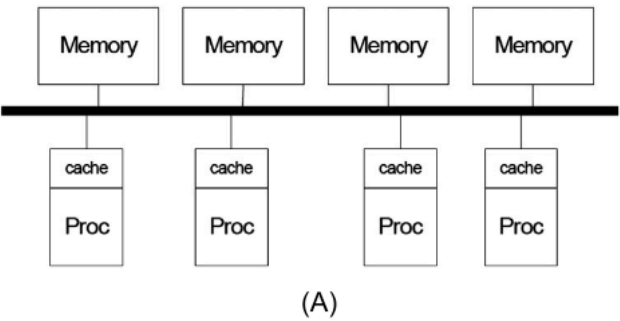
- Topology
- Routing
- Flow control

Key Network Metrics

Metric	Meaning
Node degree	Number of links attached to a node
Diameter	Longest shortest path between nodes
Average distance	Average path length between node pairs
Bisection width	Minimum number of links cut to split network in half
Bisection bandwidth	Bandwidth between two halves of the network
Cost	Switches, ports, links, cables, power

Static Topologies: Comparison

Topology	Strength	Limitation
Bus	Simple and low cost	Does not scale
Hypercube	Good bisection bandwidth	Node degree grows with $\log N$
2D mesh	Many paths, simple layout	Edge asymmetry affects performance
2D torus	Better symmetry, good for neighbor communication	Higher component cost than mesh

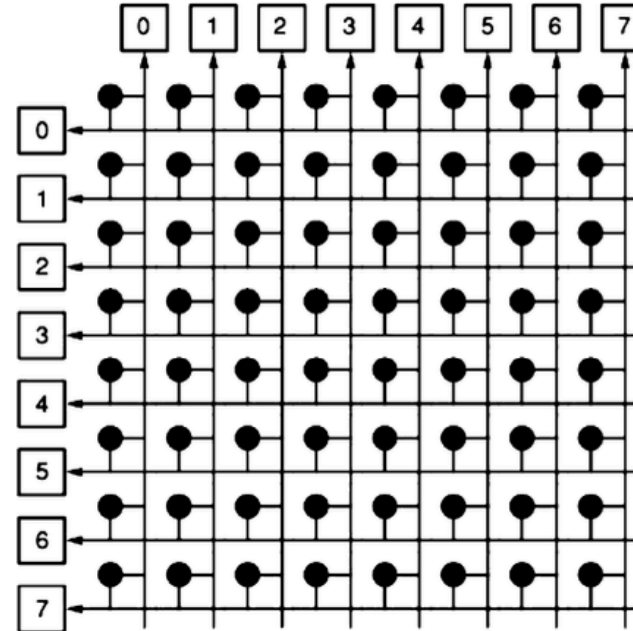


Switched Network Topologies

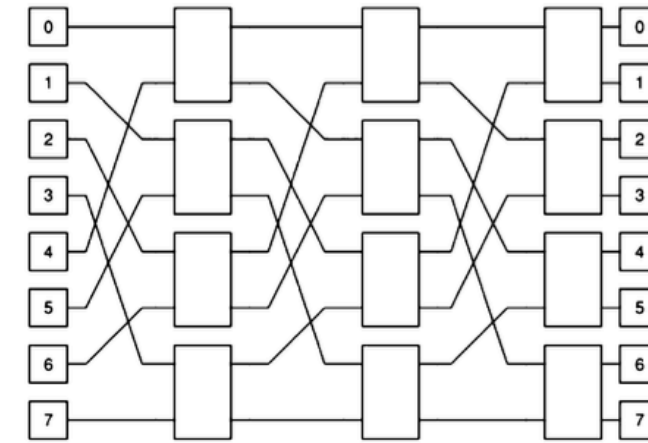
Switched networks use multiple layers of switches.

Examples:

- Crossbar
- Omega
- Butterfly
- Benes
- Banyan



(A)



(B)

Cloud Interconnection Networks

A cloud infrastructure consists of one or more warehouse-scale computers.

Typical hierarchy:

1. Servers packed into racks
2. Rack-level routers/switches
3. Cluster-level routers/switches
4. Local data center fabric
5. Inter-data-center networks

Goal:

- Low latency
- High bandwidth
- Scalable cost
- Location-transparent communication when possible

Oversubscription

Oversubscription measures whether the network can support full host bandwidth.

It is the ratio of:

- Worst-case achievable aggregate bandwidth among servers
- Total bisection bandwidth of the interconnect

Examples:

- **1:1** means any host can communicate at full bandwidth.
- **4:1** means only about 25% of server bandwidth may be available under some patterns.

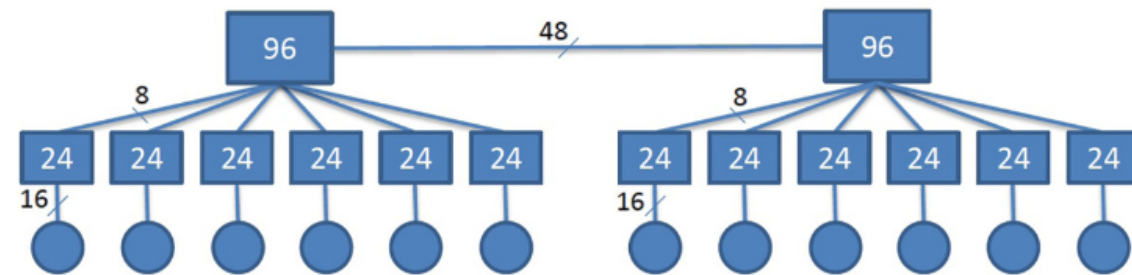
Typical values in large clusters can range from about 2.5:1 to 8:1.

Fat-Tree Networks

A fat-tree is a tree-like topology with more bandwidth near the root.

Properties:

- Servers are placed at the leaves.
- Switches populate internal nodes and root.
- Extra links near the root increase bisection bandwidth.
- Can be built from commodity switches.
- Supports many communication patterns well.

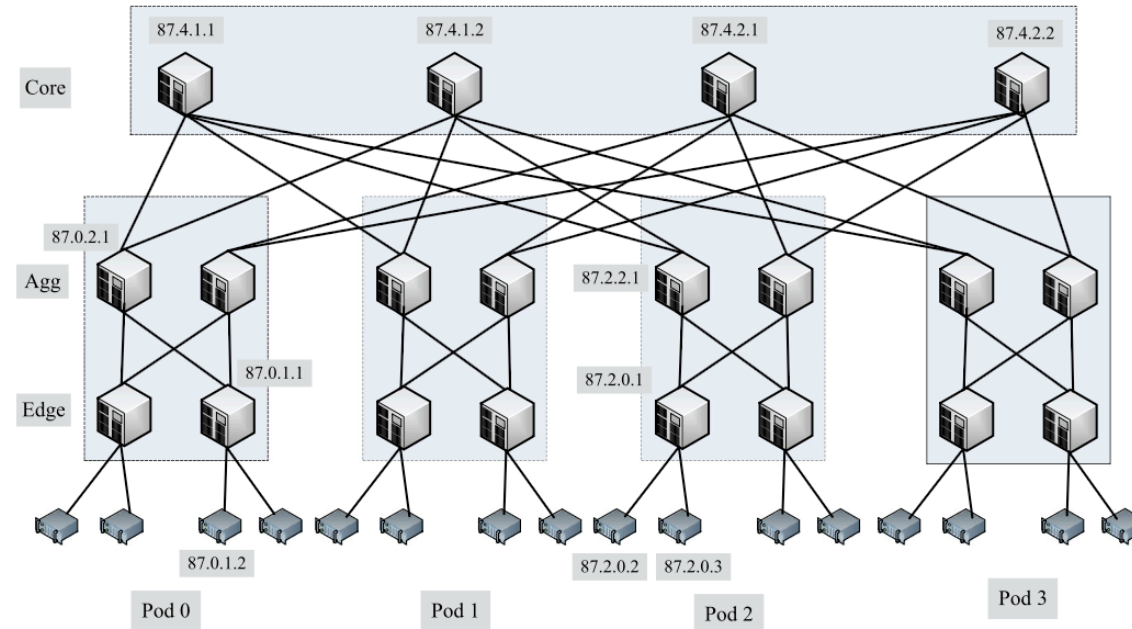


Fat-Tree Data Center Network

Fat-tree data center networks improve scalability and cost.

Design principles:

- Scale to many nodes
- Use multiple core switches
- Support multi-path routing
- Use commodity switches with good cost/performance
- Use Equal-Cost Multi-Path routing to split flows



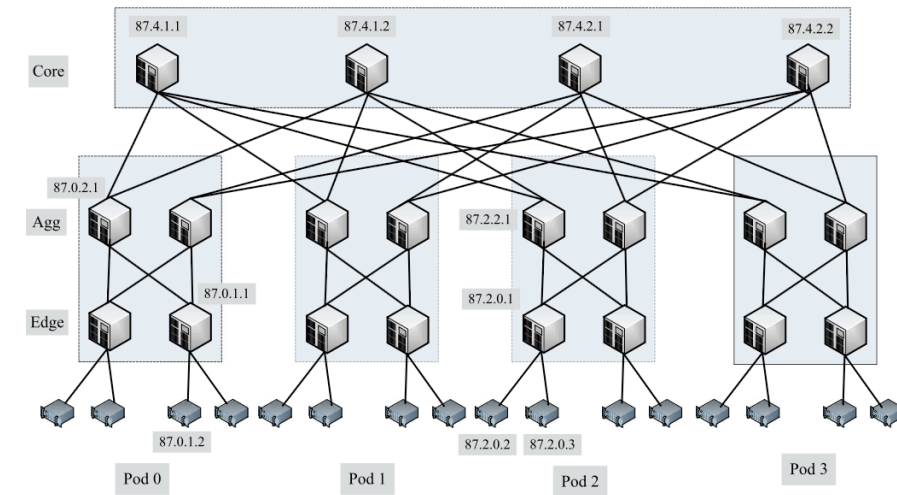
Fat-Tree with k -Port Switches

A fat-tree with k -port switches has:

- k pods
- Two switch layers per pod
- $k/2$ switches per layer in each pod
- $k/2$ servers connected to each lower-layer switch
- Multiple paths between server pairs

For a k -port design:

- Total switches: $k(k + 1)$
- Total servers: k^2
- Paths between pairs of servers: $(k/2)^2$ (diff pod), $k/2$ (same pod diff edge switch), 1 (same edge switch)



Routing in Fat-Trees

Fat-tree routing uses two-level prefix lookup.

Routing tables include:

- Main table entries: prefix to output port
- Optional secondary table: suffix to output port

Traffic diffusion occurs in the first half of the journey.

Once a packet reaches a core switch, there is a unique link toward the destination pod.

Power and Heat

Data center network design is not just about bandwidth.

Switches at higher tiers can consume several kW.

The full interconnection infrastructure may consume hundreds to thousands of kW.

Design tradeoffs include:

- Bandwidth
- Latency
- Switch cost
- Cable cost
- Power consumption
- Cooling requirements

Network Resource Management

Cloud providers must often guarantee communication bandwidth promised by SLAs.

A switch has limited:

- Link bandwidth
- Input buffer space
- Output buffer space
- CPU cycles
- Switching capacity

When overloaded, it drops packets.

Problem: A switch must schedule multiple flows fairly and efficiently.

What Must Be Managed?

For each flow, scheduling algorithms manage:

- Bandwidth allocation
- Timing of packet transmission
- Buffer space
- Fairness among flows

Simple FCFS scheduling is easy, but not fair.

Greedy flows may transmit at higher rates and consume more bandwidth.

Content Delivery Networks

Content Delivery Networks

Clouds support network-centric content as well as network-centric computing.

A Content Delivery Network (CDN):

- Receives content from an origin server
- Replicates content to edge cache servers
- Delivers content from a nearby edge server

CDNs reduce communication bandwidth and improve user experience through caching and replication.

CDN Goals

CDNs are designed to improve:

- Scalability
- Reliability
- Performance
- Security
- Capacity on demand

CDNs help absorb traffic spikes such as flash crowds.

They place resources strategically throughout the Internet.

CDN Content Types

Content Type	Description
Static content	HTML pages, images, documents, patches, audio/video files
Live streaming media	Real-time delivery from encoder to media server
On-demand streaming	Stored media such as movies or music clips

Static content is easier to cache.

Live and on-demand streaming are more challenging.

CDN Organization

Two broad approaches:

1. Overlay approach

- Network core does not actively participate.
- Content is replicated and requests are redirected.

2. Network approach

- Routers and switches use dedicated software.
- Network elements help identify application types and redirect requests.

Some CDNs use both.

CDN Design Decisions

Important CDN design choices:

1. Placement of edge servers
2. Content selection and delivery
3. Content management
4. Request routing policies

Content management includes:

- Cache maintenance
- Cache update policies
- Consistency among replicas

CDN Request Routing

A CDN directs users to an edge server based on metrics such as:

- Network proximity
- Client-perceived latency
- Distance
- Edge server load
- Available bandwidth
- Connection reliability

Common approaches:

- DNS-based routing
- HTTP redirection
- IP anycasting
- Service-node-based load balancing

CDN Performance Metrics

Metric	Meaning
Cache hit ratio	Cached objects divided by requested objects
Reserved origin bandwidth	Bandwidth protected for origin server
Latency	User-perceived response time
Edge utilization	How effectively edge servers are used
Reliability	Often based on packet loss measurements

Content-Centric Networks vs. CDNs

Content-Centric Networks and Named Data Networks generalize some CDN ideas.

CDN	CCN / NDN
Application-layer infrastructure	Network architecture idea
Edge servers cache content	Routers can cache content
Requests often routed by DNS/HTTP/IP	Requests routed by content name
Security often tied to connection/channel	Security tied to content signature

Summaries

- Cloud performance is deeply tied to network latency, bandwidth, and topology.
- The Internet provides broad cloud access but only best-effort packet delivery.
- TCP improves reliability and fairness but affects cloud and web latency.
- NDN and SDN represent alternative ways to rethink networking.
- Data center networks must balance bandwidth, cost, power, and scalability.
- CDNs move content closer to users to improve performance and reduce bandwidth.